

Scalar k:2k four-moment Radau all-amplitude closure

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

R-115 closes the original R-112 stationary scalar $k:2k$ log-Laplace inequality for every covariance shape, every amplitude, and every $\tau \geq 0$. R-114 already covers $0 \leq b \leq 643/200$. On the complementary half-line $b \geq 643/200$, a degree-four left Gauss–Radau majorant transfers the original Laplace transform to a three-atom law. An exact three-atom skew lemma and a finite outward-rounded Arb cover prove that the tilted variance of that law never exceeds its initial variance. The covariance reserve then gives a strict gap for $\tau > 0$ and equality only at $\tau = 0$.

This is a complete theorem for the stationary scalar model, not yet a theorem for the production A1 tensor. The full one-fresh-root owner-preserving A1 embedding, all visits and inputs, random- W double divergence, complete heat, mixed baseline, trace and rational recovery, and the separate one-use source/sextic aggregation remain open. Consequently $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector A remain open. R-103 REG and R-087 fixed-cutoff CORE keep their accepted owners. Historical R-085 weighted/labelled-owner formulas remain non-load-bearing.

All displayed algebraic identities are exact. The primary executable reconstructs the packet from its phase/exponential definition and runs a finite directed-rounding cover. A structurally independent executable reconstructs the same moments and replaces that cover by an exact Bernstein/radical certificate. Ordinary floating-point scans are used only to choose adversarial fixtures and are not theorem evidence.

1 Exact scalar packet and inherited cone

Retain the R-112 variables

$$c = \frac{v}{v+4w}, \quad s = 1 - c, \quad b = \frac{A^2}{v+4w}, \quad \tau = q(v+4w)^2, \quad x = b\tau. \quad (1.1)$$

Let T, U be independent unit exponentials, let ϕ be uniform, and set

$$\rho = \frac{cT}{2}, \quad \sigma = \frac{sU}{8}, \quad (1.2)$$

$$L = \rho + 4\sigma - \frac{1}{2}, \quad Q = \rho^2 + 10\rho\sigma + 4\sigma^2 - \rho - \sigma, \quad B = 6\rho\sqrt{\sigma} \cos \phi. \quad (1.3)$$

The exact centered packet and covariance square are

$$F_b = bL + \sqrt{b}B + Q, \quad \mathbb{E}F_b = 0, \quad (1.4)$$

$$K = b^2K_2 + bK_1 + K_0, \quad (1.5)$$

where

$$K_2 = c^2 - c + \frac{1}{2}, \quad K_1 = \frac{-3c^3 + 5c^2 + c + 1}{4}, \quad (1.6)$$

$$K_0 = \frac{153c^4 - 156c^3 + 82c^2 - 4c + 5}{64}. \quad (1.7)$$

Its variance has the exact reserve

$$V := \text{Var}(F_b) = \frac{K - \Delta}{2}, \quad \Delta = \frac{33c^4 - 36c^3 + 22c^2 - 4c + 1}{64} \geq \frac{1}{100}. \quad (1.8)$$

The target is

$$\boxed{\log \mathbb{E} e^{-\tau F_b} \leq \frac{K\tau^2}{4}}. \quad (1.9)$$

R-114 proves (1.9), strictly for $\tau > 0$, on the complete closed cone $0 \leq b \leq 643/200$. It also proves the sharp support floor

$$\boxed{F_b \geq -\frac{b}{2} \quad (b \geq 6/5)}, \quad (1.10)$$

with equality only at $\rho = \sigma = 0$. Thus the complementary proof may set

$$\beta = \frac{b}{2}, \quad Z = F_b + \beta \geq 0. \quad (1.11)$$

2 Four exact moments

Write $\mu_j = \mathbb{E} F_b^j$. Phase symmetry and $\mathbb{E} \cos^{2j} \phi = \binom{2j}{j}/4^j$ give

$$\mu_n = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \frac{\binom{2j}{j}}{4^j} \mathbb{E} [(bL + Q)^{n-2j} (36b\rho^2\sigma)^j]. \quad (2.1)$$

The exponential moments $\mathbb{E} T^i U^j = i!j!$ make (2.1) a finite exact polynomial. In particular,

$$32\mu_2 = 16b^2c^2 - 16b^2c + 8b^2 - 12bc^3 + 20bc^2 + 4bc + 4b + 30c^4 - 30c^3 + 15c^2 + 1, \quad (2.2)$$

and $\mu_2 = V$. The third moment is

$$\begin{aligned} 256\mu_3 = & 192b^3c^2 - 192b^3c + 64b^3 + 48b^2c^4 - 240b^2c^3 + 672b^2c^2 - 192b^2c + 96b^2 \\ & + 1404bc^5 - 1860bc^4 + 912bc^3 + 552bc^2 - 108bc + 60b \\ & + 990c^6 + 108c^5 - 375c^4 + 66c^3 + 186c^2 - 30c + 15. \end{aligned} \quad (2.3)$$

For later independent reconstruction, the fourth moment is

$$\begin{aligned} \mu_4 = & \frac{3b^4}{16}(8c^4 - 16c^3 + 20c^2 - 12c + 3) \\ & - \frac{3b^3}{16}(18c^5 - 62c^4 + 61c^3 - 61c^2 + 23c - 7) \\ & + \frac{3b^2}{64}(480c^6 - 558c^5 + 262c^4 + 10c^3 + 373c^2 - 102c + 31) \\ & + \frac{3b}{128}(99c^7 + 4041c^6 - 4248c^5 + 2068c^4 - 107c^3 + 543c^2 - 128c + 36) \\ & + \frac{3}{512}(12474c^8 - 9486c^7 + 9213c^6 - 4788c^5 + 1768c^4 - 518c^3 + 661c^2 - 144c + 36). \end{aligned} \quad (2.4)$$

No fifth moment is used. As a useful boundary check, put $d = 1/b = (200/643)y$. The tensor Bernstein expansion of $256d^3\mu_3$ on $(y, c) \in [0, 1]^2$ has 28/28 positive rational coefficients, with minimum $32/5$. Therefore

$$\boxed{\mu_3(F_b) \geq \frac{b^3}{40} \quad (b \geq 643/200).} \quad (2.5)$$

This proves that the original tilted variance initially decreases. It is not the all-tilt proof: at $(b, c) = (643/200, 1/2)$ the fourth cumulant is $4376252798439/51200000000 > 0$.

The shifted moments needed below are

$$m_1 = \beta, \quad m_2 = \beta^2 + \mu_2, \quad m_3 = \beta^3 + 3\beta\mu_2 + \mu_3, \quad (2.6)$$

$$m_4 = \beta^4 + 6\beta^2\mu_2 + 4\beta\mu_3 + \mu_4. \quad (2.7)$$

3 The sharp four-moment Gauss–Radau majorant

Lemma 3.1 (left Radau majorant)

Let $Z \geq 0$ have finite fourth moment and a nondegenerate Stieltjes moment matrix. Define

$$D_0 = m_1m_3 - m_2^2, \quad A_0 = \frac{m_1m_4 - m_2m_3}{D_0}, \quad B_0 = \frac{m_2m_4 - m_3^2}{D_0}. \quad (3.1)$$

Let $0 < r_1 < r_2$ be the roots of $z^2 - A_0z + B_0$. There are unique positive weights w_0, w_1, w_2 on $0, r_1, r_2$ matching $m_0, \dots, m_4$; explicitly

$$w_1 = \frac{m_1r_2 - m_2}{r_1(r_2 - r_1)}, \quad w_2 = \frac{m_2 - m_1r_1}{r_2(r_2 - r_1)}, \quad w_0 = 1 - w_1 - w_2. \quad (3.2)$$

Then, for every $t \geq 0$,

$$\boxed{\mathbb{E}e^{-tZ} \leq w_0 + w_1e^{-r_1t} + w_2e^{-r_2t}.} \quad (3.3)$$

Proof. The polynomial $z^2 - A_0z + B_0$ is the monic degree-two orthogonal polynomial for the positive measure $z d\Pr_Z$. Hence its roots lie in $(0, \infty)$. The Radau rule is exact through degree four, and each quadrature weight is positive because it equals the expectation of the square of its cardinal polynomial. Let H_t be the degree-four Hermite interpolant of e^{-tz} , matching once at zero and twice at each r_i . Its remainder is

$$e^{-tz} - H_t(z) = \frac{-t^5 e^{-t\xi_z}}{5!} z(z - r_1)^2(z - r_2)^2 \leq 0. \quad (3.4)$$

Taking expectations and using degree-four exactness proves (3.3). The bound is moment-sharp: the three-atom Radau law has the same four moments and attains equality. $\square$

Applied to (1.11), Lemma 3.1 gives

$$\mathbb{E}e^{-tF_b} \leq e^{\beta t} \{w_0 + w_1e^{-r_1t} + w_2e^{-r_2t}\}. \quad (3.5)$$

4 A three-atom all-tilt skew lemma

Write

$$v = r_1, \quad u = r_2 - r_1, \quad a = w_0, \quad q = w_1, \quad p = w_2. \quad (4.1)$$

Lemma 4.1 (sufficient nonnegative skew)

Assume

$$u \geq v > 0, \quad p \leq \min(a, q). \quad (4.2)$$

If $q \leq a$, the third centered moment of the law on $0, v, v + u$ remains nonnegative under every negative exponential tilt. If $q > a$, put

$$k = \frac{u}{v}, \quad h = \frac{kq}{(k+1)a}, \quad (4.3)$$

$$F = \begin{cases} (q-a)/p, & h \leq 1, \\ \frac{q}{p(k+1)}h^k, & h \geq 1, \end{cases} \quad (4.4)$$

and

$$\Phi = (k-1)(k+2)(2k+1) + k^3 \frac{q-p}{q} + (k+1)^3 \frac{a-p}{q} - F. \quad (4.5)$$

Then $\Phi \geq 0$ is sufficient for the same all-tilt conclusion.

Proof. Under the tilt, write

$$A = a, \quad B = qe^{-vt}, \quad C = pe^{-(u+v)t}, \quad S = A + B + C.$$

Direct expansion gives

$$\begin{aligned} S^3 \mu_{3,t} &= AB(A-B)v^3 + AC(A-C)(u+v)^3 + BC(B-C)u^3 \\ &\quad + ABC(u-v)(u+2v)(2u+v). \end{aligned} \quad (4.6)$$

If $q \leq a$, every term is nonnegative. Otherwise the only adverse interval has $B > A$. On that interval, division by ABC and (4.2) give

$$\frac{S^3 \mu_{3,t}}{ABCv^3} \geq -\frac{B-A}{C} + k^3 \frac{q-p}{q} + (k+1)^3 \frac{a-p}{q} + (k-1)(k+2)(2k+1). \quad (4.7)$$

Indeed $B > A$ implies $e^{-vt} > a/q$, which yields $(B-C)/A \geq (q-p)/q$; the other pair is bounded similarly. With $s = e^{vt} \in [1, q/a]$,

$$\frac{B-A}{C} = \frac{(q-as)s^k}{p}. \quad (4.8)$$

Its derivative has the sign of $kq - (k+1)as$, so its exact maximum is (4.4). Equations (4.5)–(4.8) prove the claim. $\square$

If $\Psi(t)$ is the logarithm of the right side of (3.5), then

$$\Psi''(t) = \text{Var}_t(Z), \quad \Psi'''(t) = -\mu_{3,t}(Z). \quad (4.9)$$

Thus Lemma 4.1 makes Ψ'' nonincreasing.

5 Finite exact and outward-rounded certificate

Put $d = 1/b$ and divide m_j by b^j . Every scaled moment is an exact polynomial in (d, c) , including the endpoint $d = 0$. The residual half-line becomes the compact rectangle

$$\boxed{0 \leq d \leq \frac{200}{643}, \quad 0 \leq c \leq 1.} \quad (5.1)$$

The executables reconstruct the degree-two Radau polynomial from (2.1), not from hardcoded derived nodes or weights.

For an exact algebraic cross-check, let H, N_A, N_B be the coefficients of the two free nodes in the endpoint coordinate $X = -F_b \leq b/2$:

$$\begin{aligned} H &= \beta^2 \mu_2 + \beta \mu_3 - \mu_2^2, \\ N_A &= \beta^2 \mu_3 - \beta \mu_2^2 + \beta \mu_4 - \mu_2 \mu_3, \\ N_B &= -\beta^2 \mu_2^2 - \beta \mu_2 \mu_3 + \mu_2 \mu_4 - \mu_3^2. \end{aligned} \quad (5.2)$$

Set $D = N_A^2 - 4HN_B$ and $J = 2H\beta + N_A$. Exact tensor-Bernstein tables prove

$$H > 0, \quad D > 0, \quad J^2 - D > 0, \quad 9D - J^2 > 0, \quad (5.3)$$

so $r_1 > 0$ and $r_2 > 2r_1$. Further exact radical reductions prove

$$p < q, \quad p < a, \quad p \geq \frac{1}{200}, \quad q \geq \frac{15}{32}, \quad a \geq \frac{1}{6}, \quad 2 \leq k \leq 6, \quad kq \leq 9a, \quad h \leq 3. \quad (5.4)$$

The $p \leq a$ proof needs only five de Casteljau leaves; no box is unresolved. These polynomial checks independently protect every denominator and radical used by the interval proof. The independent engine reconstructs 21 structural polynomials and certifies them on 24 exact leaves: 8298 rational coefficients consist of 8286 strict positives and 12 retained endpoint zeros. Its disjunctive $p \leq a$ cover has five leaves and 4231/4231 positive coefficients. Thus no prerequisite is imported from the Arb engine.

The same exact path also closes the remaining skew margin. Indeed, for

$$g(k) = \frac{1}{k+1} \left(\frac{9}{k+1} \right)^k,$$

both branches of (4.4), using $kq \leq 9a$, obey

$$F \leq \frac{q}{p} g(k), \quad \max_{2 \leq k \leq 6} g(k) = \frac{e^{9/e}}{9} < \frac{25}{8}. \quad (5.5)$$

The first inequality on the $h \leq 1$ branch follows from $q - a \leq q/(k+1)$; on the $h \geq 1$ branch it follows directly from $h \leq 9/(k+1)$. The derivative of $\log g$ is $\log(9/(k+1)) - 1$, and elementary directed interval bounds certify the final strict comparison in (5.5).

Replace F in (4.5) by $25q/(8p)$ and call the smaller margin $\bar{\Phi}$. After clearing the already certified positive denominator, the independent engine reduces its numerator to $A + B\sqrt{D}$. On the whole rectangle (5.1), exact tensor-Bernstein tables prove $A > 0$, $-B > 0$, and $A^2 - DB^2 > 0$ with respectively 2850/2850, 2145/2145, and 11175/11175 positive rational coefficients. Hence $A + B\sqrt{D} > 0$, so

$$\boxed{\Phi \geq \bar{\Phi} > 0} \quad (5.6)$$

without interval subdivision.

The load-bearing transcendental check uses python-flint Arb balls. The primary implementation starts from a deterministic 16×32 rational grid on (5.1), evaluates exact scaled-moment polynomials by Horner arithmetic, and propagates first-order centered forms, recentered after every operation. Each leaf must prove all inequalities in (4.2), and then either $q \leq a$, or the applicable branch of $\Phi > 0$. A leaf crossing $h = 1$ must prove both branches. Failed or nonfinite leaves are bisected; none is discarded.

The structurally different primary directed-rounding cover has

$$\boxed{46714 \text{ evaluations,} \quad 23613 \text{ accepted leaves,} \quad 0 \text{ pending leaves.}} \quad (5.7)$$

Its weakest outward lower enclosure for an accepted Φ branch is

$$\boxed{\Phi > 0.0019 > 0.} \quad (5.8)$$

The exact independent engine and the primary Arb engine share no source imports and are hash-pinned separately by the integrated verifier.

Theorem 5.1 (all-amplitude scalar closure)

For every $b \geq 0$, $c \in [0, 1]$, and $\tau \geq 0$,

$$\boxed{\log \mathbb{E} e^{-\tau F_b} \leq \frac{K(c, b)\tau^2}{4}}. \quad (5.9)$$

Equality holds at $\tau = 0$; the inequality is strict for $\tau > 0$.

Proof. R-114 covers $0 \leq b \leq 643/200$. On $b \geq 643/200$, the finite certificate proves Lemma 4.1 for the Radau law. Hence (4.9) and moment matching give

$$\Psi''(t) \leq \Psi''(0) = \text{Var}(F_b) = V = \frac{K - \Delta}{2} < \frac{K}{2}. \quad (5.10)$$

Since $\Psi(0) = \Psi'(0) = 0$, two integrations yield $\Psi(t) \leq Vt^2/2 < Kt^2/4$ for $t > 0$. Lemma 3.1 transfers this upper bound to the original packet. $\square$

6 Auxiliary two-moment endpoint and exact method failures

The proof search also sharpened the R-114 two-moment boundary. For $\beta = b/2$, set

$$A = \beta^2 + V, \quad D_1 = \beta^2 - V, \quad y = \frac{D_1}{A}. \quad (6.1)$$

The self-contained lower bound

$$\frac{\text{atanh } y}{y} \geq 1 + \frac{y^2}{3} + \frac{y^4}{5}, \quad (6.2)$$

follows because the derivative of $\text{atanh } y - y - y^3/3 - y^5/5$ is $y^6/(1 - y^2)$. The cleared sufficient polynomial is

$$T_4 = 30K\beta^2 A^4 + 10K\beta^2 D_1^2 A^2 + 6K\beta^2 D_1^4 - 15A^6. \quad (6.3)$$

Seven rational cells give 1655/1655 positive exact Bernstein coefficients, extending that route to

$$\frac{643}{200} \leq b \leq \frac{103}{32}. \quad (6.4)$$

This extension is independent confirmation, but Theorem 5.1 no longer needs it.

The exact Kearns–Saul coefficient itself fails immediately afterward. At

$$b = \frac{3219}{1000}, \quad c = \frac{31}{100}, \quad y = \frac{7333898609}{96285711391}, \quad (6.5)$$

the variance branch is unavailable and

$$\frac{\text{atanh } y}{y} \leq 1 + \frac{y^2}{3} + \frac{y^4}{5(1 - y^2)}. \quad (6.6)$$

Substitution makes the exact KS margin at most

$$-\frac{72442776419046601199446847233957478399499392897}{3928811407925749185840216970677658361040000000000000000} < 0. \quad (6.7)$$

Its sharp equality time lies below $1/8$, while the live compact cap exceeds 2. Thus the failure occurs in the relevant time range. It retires only every proof using the support and first two moments through the exact KS constant; it is not a target counterexample.

Nor do support, four moments, and a positive covariance reserve imply the target for an arbitrary law. Let $X = -Y$ take values $-1, 0, 2$ with weights $1/2, 1/4, 1/4$. Then $Y \geq -2$, $\text{Var}(Y) = 3/2$, and $K = 16/5 > 3$. At $t = 1/2$,

$$\mathbb{E}e^{tX} = \frac{e^{-1/2}}{2} + \frac{1}{4} + \frac{e}{4} > e^{1/5}. \quad (6.8)$$

Indeed $e^{-1/2} > 29/48$, $e > 1957/720$, and $e^{1/5} < 61/50 + 2/1425$, leaving the rational gap $2789/273600 > 0$. This proves that the actual-packet inequalities in Section 5, rather than a blanket four-moment principle, are load-bearing.

The primary SymPy and independent standard-library `Fraction` method-boundary children each reconstruct all 1655/1655 extension signs, the exact KS failure margin and live-time comparison, and the rational gap in (6.8). Neither child imports either Radau implementation.

Finally, the corrected symmetric Bessel route remains reusable. With $\sigma = (1 - c)U/8$ and suitable A, B_0, δ, C_0 , exact integration in T reduces the cancellation-preserving surrogate to

$$\int_0^\infty e^{-U-\tau C_0} \left\{ \frac{J_0(A, B_0)}{2} + \frac{J_0(A, B_0 - \delta) + J_0(A, B_0 + \delta)}{4} - \frac{\delta^4 J_4(A, B_0)}{192} \right\} dU, \quad (6.9)$$

where $2AJ_{n+1} + B_0J_n = nJ_{n-1}$. It is no longer needed for the scalar closure and is parked for possible full-A1 comparison.

7 Proof-evidence roadmap

The successful path and the retired branches are now separated as follows:

1. R-110 identifies the exact stationary scalar packet and local reserve. R-111 closes both covariance faces. R-112 compactifies the mixed problem, R-113 makes its boundaries effective, and R-114 closes $b \leq 643/200$.
2. The fifth-order atanh term gives a small exact cone extension, but the full KS witness (6.7) prevents further two-moment work from being useful.
3. Four-moment Radau is the correct moment-sharp replacement. A blanket four-moment reserve theorem is false by (6.8), so the proof retains the actual packet's node, weight, and skew geometry.
4. Lemma 4.1 compresses the all-time problem to one two-parameter function. The finite outward cover of (5.1) proves that function positive and closes the stationary scalar theorem.
5. The next proof object is not another scalar classifier. It is the full A1 one-fresh-root, contraction-closed conditional owner cluster, followed by the separate one-use source/sextic aggregation. No stationary scalar result may silently replace that embedding.

8 Devil's-advocate review

1. **Objection:** The quadrature rule approximates the original Laplace transform without a one-sided error. **Verdict: DISMISSED.** The Hermite remainder (3.4) has a fixed nonpositive sign on $Z \geq 0$, and the Radau rule is exact through degree four. The direction in (3.3) is analytic, not numerical.
2. **Objection:** Positivity at sampled (b, c, t) points was promoted to an all-time theorem. **Verdict: DISMISSED.** Lemma 4.1 maximizes the only adverse exponential ratio over every $t \geq 0$ exactly. The remaining (d, c) rectangle is covered by outward Arb balls with zero pending leaves.

3. **Objection:** Interval subdivision discarded nonfinite or nonpositive cells. **Verdict: DISMISSED.** Every failed leaf is bisected and retained. Completion requires an empty pending queue. Exact Bernstein geometry independently protects the radicals, weights, and denominators.
4. **Objection:** The old global tilted-variance counterexample refutes this proof. **Verdict: DISMISSED.** That counterexample applies to a different packet. R-115 proves nonnegative skew for the three-atom Radau law of the actual R-112 packet using the packet-specific certificate (5.1)–(5.6); no blanket monotonicity assertion is made.
5. **Objection:** Four moments and $K > 2V$ alone prove the result. **Verdict: UPHELD AS FALSE.** The exact law (6.8) violates that claim. The actual-packet skew inequalities are essential and are pinned as evidence.
6. **Objection:** Scalar closure proves the A13 one-use theorem or Sector A. **Verdict: UPHELD AS A FORBIDDEN READING.** The stationary scalar model has not yet been embedded into the full A1 owner tensor, and the separate one-use source/sextic aggregation is unproved. Tier stays T4.

9 Result footer

Result ID: A13-CLASSII-SCALAR-K2K-FOUR-MOMENT-RADAU-ALL-AMPLITUDE-CLOSURE

Precise statement: R-115 proves the original R-112 stationary scalar k:2k

log-Laplace inequality for every covariance shape c in $[0,1]$, every amplitude $b \geq 0$, and every $\tau \geq 0$, strictly for $\tau > 0$. R-114 covers $b \leq 643/200$; a four-moment Radau majorant, exact three-atom skew lemma, exact algebraic guards, and a complete outward Arb cover close $b \geq 643/200$.

Scope: The original stationary scalar model only; not the full A1 tensor embedding or the one-use source/sextic aggregation.

Dependencies: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION;

R-087; R-103; R-105; R-110--R-114.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command:

E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/

a13_classii_scalar_k2k_four_moment_radau_all_amplitude_verify.py

Expected output: Integrated R-115 PASS; primary and independent children

PASS. The primary enforces 46714 evaluations, 23613 accepted leaves, zero pending leaves, and a positive outward margin. The independent child enforces the 2850, 2145, and 11175 exact positive Bernstein counts and their canonical hashes. Two method-boundary children independently enforce 1655/1655 extension signs, the exact KS failure inside the live time range, and the generic reserve-only gap 2789/273600.

Falsification gate: Any failed moment, Radau/Hermite identity, skew identity, exact geometry guard, directed-rounding leaf, child/source/note/PDF hash, nonempty pending queue, or promotion beyond the stationary scalar scope.

Tier before / after: T4 / T4.

No-overclaim statement: No full A1 owner-preserving embedding, adapted production cluster, one-use source/sextic aggregation, OVERLAP_src, Nelson bound, removal limit, interacting measure, Sector-A closure, or higher tier is established.

Next required action: Embed only this proved scalar normalizer into the full A1 one-fresh-root contraction-closed conditional owner cluster, retaining complete heat, mean, trace, mixed baseline, rational recovery, every visit, random-W double divergence, and the R-063 forest exactly once; then prove the separate one-use source/sextic aggregation.